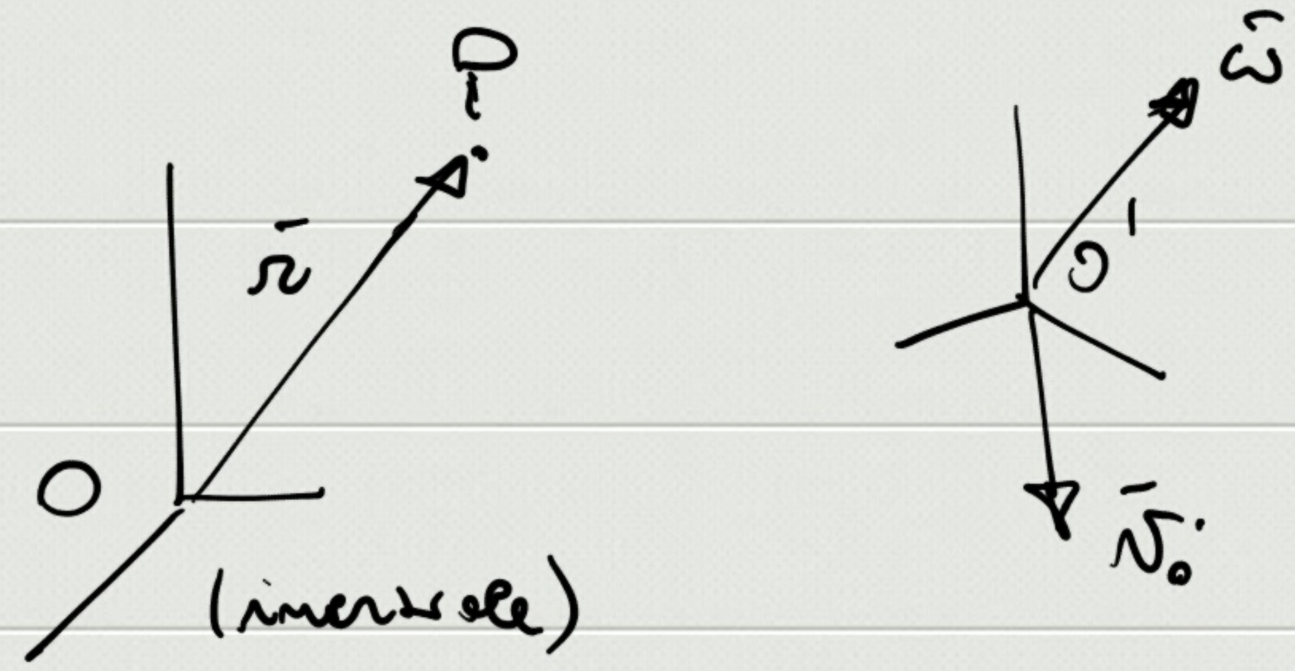
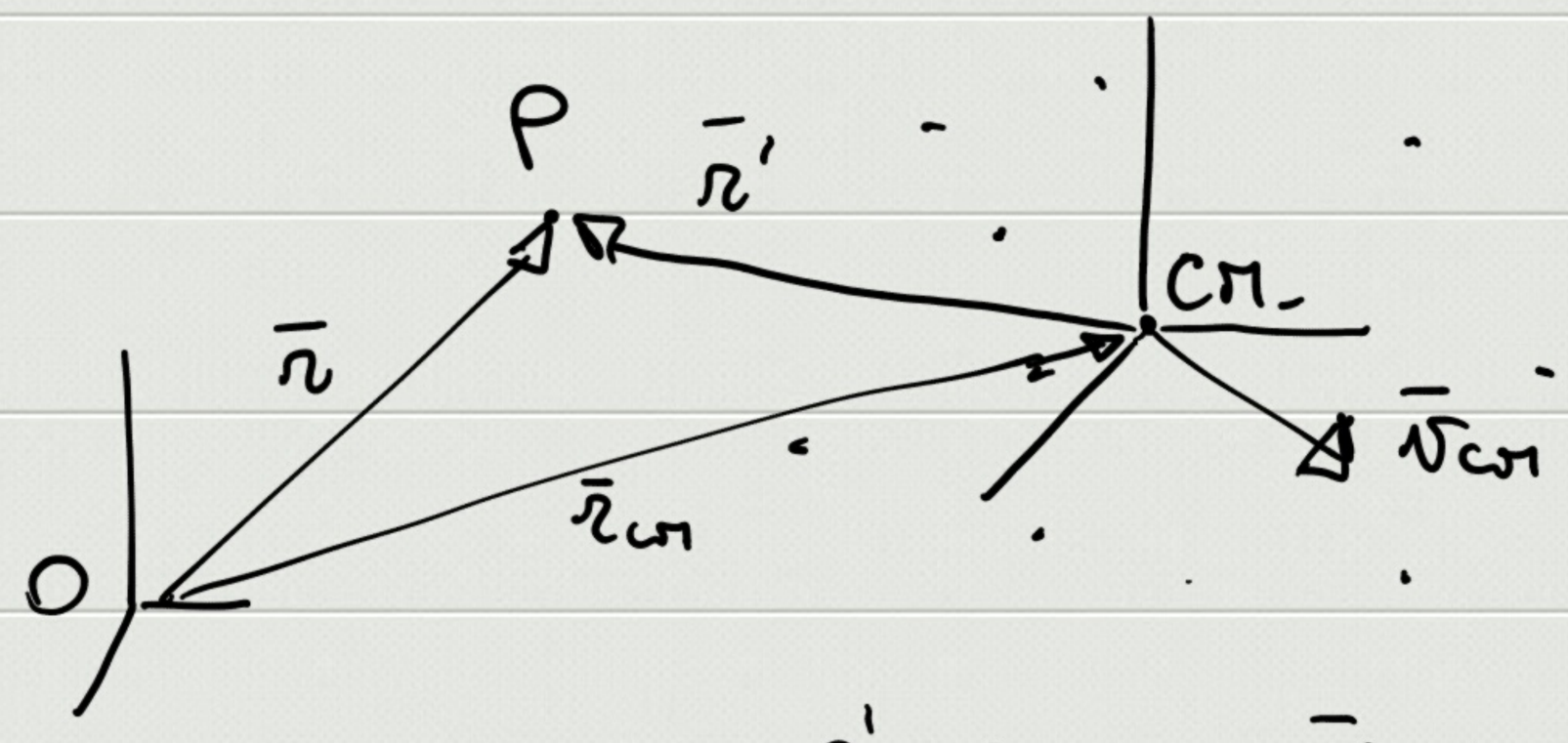



$$\bar{r} = \bar{r}' + \bar{r}_{O'}$$

$$\bar{v} = \bar{v}' + \bar{v}_{O'} + \bar{\omega} \times \bar{r}'$$

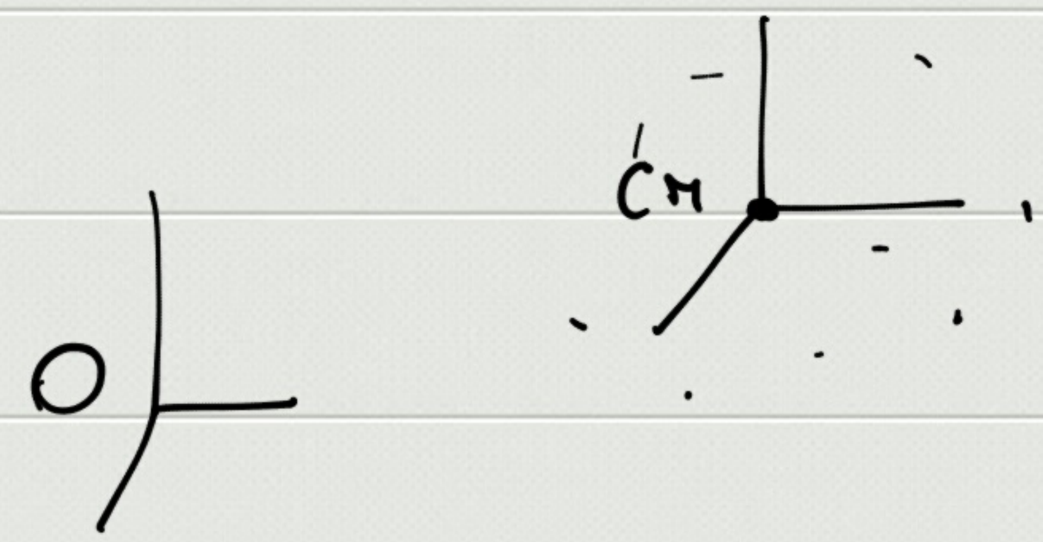
$$\bar{a} = \bar{a}' + \bar{a}_{O'} + \bar{\omega} \times (\bar{\omega} \times \bar{r}') + \frac{d\bar{\omega}}{dt} \times \bar{r}' + 2\bar{\omega} \times \bar{v}'$$




$O' \equiv CM \qquad \bar{\omega} = 0$

Sistema di riferimento del centro di massa

$$\begin{cases} \bar{r} = \bar{r}' + \bar{r}_{CM} \\ \bar{v} = \bar{v}' + \bar{v}_{CM} \\ \bar{a} = \bar{a}' + \bar{a}_{CM} \end{cases} \Rightarrow \begin{cases} \bar{r}' = \bar{r} - \bar{r}_{CM} \\ \bar{v}' = \bar{v} - \bar{v}_{CM} \\ \bar{a}' = \bar{a} - \bar{a}_{CM} \end{cases}$$




$$\left.\begin{aligned} \bar{r}'_{CM} &= 0 \\ &\overset{!}{=} \frac{\sum_i m_i \bar{r}'_i}{\sum_i m_i} \end{aligned}\right\} \Rightarrow \boxed{\sum_i m_i \bar{r}'_i = 0}$$

$$\left.\begin{aligned} \bar{v}'_{CM} &= 0 \\ &\overset{!}{=} \frac{\sum_i m_i \bar{v}'_i}{\sum_i m_i} \end{aligned}\right\} \boxed{\sum_i m_i \bar{v}'_i = 0}$$

$$\bar{P}' = \sum_i \bar{p}'_i = \sum_i m_i \bar{v}'_i = 0 \Rightarrow \boxed{\bar{P}' = 0}$$

$$\bar{P}' = \sum_i m_i \bar{v}'_i = \sum_i m_i (\bar{v}_i - \bar{v}_{CM}) =$$

$$= \sum_i m_i \bar{v}_i - \sum_i m_i \bar{v}_{CM} = \bar{P} - m_{TOT} \bar{v}_{CM} = 0$$

$$\Rightarrow \bar{P} = m_{TOT} \bar{v}_{CM}$$

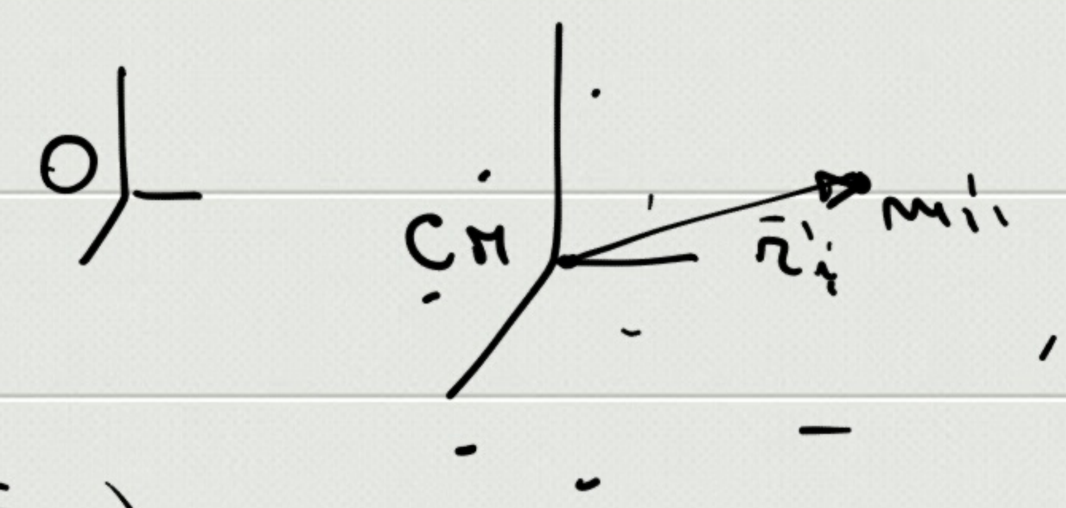
$$\left.\begin{aligned} \bar{a}'_{cm} &= 0 \\ &\overset{!}{=} \frac{\sum_i m_i \bar{a}'_i}{\sum_i m_i} \end{aligned}\right\} \Rightarrow \boxed{\sum_i m_i \bar{a}'_i = 0}$$

$$\bar{R}' = \sum_i \bar{F}'_i = \sum_i m_i \bar{a}'_i = 0 \Rightarrow \boxed{\bar{R}' = 0}$$

$$\bar{R}' = \sum_i m_i \bar{a}'_i = \sum_i m_i (\bar{a}_i - \bar{a}_{cm}) = \sum_i m_i \bar{a}_i - \sum_i m_i \bar{a}_{cm} =$$

$$= \sum_i \bar{F}_i - m_T \bar{a}_{cm} = 0 \Rightarrow \boxed{\bar{R}^E = m_T \bar{a}_{cm}}$$

$$\bar{L}_{cm} = \sum_i \bar{L}_{i,cm} =$$




$$= \sum_i \bar{r}'_i \times m_i \bar{v}_i =$$

$$= \sum_i \bar{r}'_i \times m_i (\bar{v}'_i + \bar{v}_{cm}) =$$

$$= \sum_i \bar{r}'_i \times m_i \bar{v}'_i + \sum_i \bar{r}'_i \times m_i \bar{v}_{cm} =$$

$$= \sum_i \bar{L}'_{cm,i} + \underbrace{\cancel{\sum_i m_i \bar{r}'_i}}_{=0} \times \bar{v}_{cm} =$$

$$= \bar{L}'_{cm}$$

$$\Rightarrow \boxed{\bar{L}_{cm} = \bar{L}'_{cm}}$$

$$\bar{M}^{E}_{CM} = \bar{M}'^{E}_{CM}$$

$$\Rightarrow \text{pole CM} : \quad \frac{d\bar{L}_0}{dt} = -\bar{v}_0 \times \cancel{M_{tot}\,\bar{v}_{CM}} + \bar{M}^{E}_{0}$$

$$\Rightarrow \frac{d\bar{L}_{CM}}{dt} = \bar{M}^{E}_{CM}$$

$$\Rightarrow \boxed{\frac{dL'_{CM}}{dt} = \bar{M}'^{E}_{CM}}$$